



Checking understanding of sequences

Task A: Sequences with a constant additive pattern

1) These pupils are making number patterns from a rule



Aisha



Andeep



Izzy



Jacob



Jun



Lucas



Laura



Sam



Alex



Sofia

Aisha says any number, Andeep adds a number to it. Each pupil then adds the same number Andeep added. For each question, justify your answer.

a) If Aisha says 6 and Andeep says 10, what will Izzy say?

Izzy will say 14 as Andeep added 4 to Aisha's number and $10 + 4 = 14$

b) If Aisha says 6 and Andeep says 10, what will Sofia say?

Sofia will say 42 as 4 will have been added another 8 times after Andeep says 10

c) After a different starting number Sam says 17 and Alex says 19. What did Aisha say?

Aisha said 3. Each pupil is adding 2. Subtracting 2 seven times from 17 is 3

d) If Aisha says 4 to start and Jun says 40 What will Lucas say?

Lucas says 49. A number has to be added 4 times to get from Aisha to Jun. The difference between their numbers is 36 so each pupil adds 9

e) Aisha starts with 10, only Lucas' number is another integer, what could Andeep have said?

Andeep could have added any number which is a multiple of 0.2

e.g $10 + 0.4 = 10.4$ or $10 + 5.6 = 15.6$ or $10 + 3.2 = 13.2$

There are 5 additions between Aisha and Lucas if Lucas said 11 then

$1 \div 5 = 0.2$

If Lucas said 12 then $2 \div 5 = 0.4$



Task B: Sequences with a constant multiplicative pattern

1) Sort these sequences into the correct column

Sequences with a constant additive relationship	Sequences with a constant multiplicative relationship
5, 10, 15, 20, ...	1, 2, 4, 8, ...
5.7, 8.1, 10.5, 12.9, ...	3, 30, 300, 3000, ...
1.5, 0.5, -0.5, -1.5, ...	48, 24, 12, 6, ...
6, 48, 90, 132, ...	3, -6, 12, -24, ...
	0.5, 1.5, 4.5, 13.5, ...

2) Find the missing numbers in these sequences with constant multiplicative patterns.

a) 6, 12, 24, 48, ...

e) 7, 21, 63, 189, ...

b) 5, 30, 180, 1080, ...

f) 1, 0.2, 0.04, 0.008, ...

c) 160, 80, 40, 20, ...

g) 0.4, 8, 160, 3200, 64 000, ...

d) 0.3, -3, 30, -300, ...

h) 4.5, 6, 8, $\frac{32}{3}$, ...

